Extended experiments in volumetric fluorescence recovery

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FRIS research record

Evidence snapshot: 12 October 2026

Experimental directions, findings and conclusions beyond the main manuscript

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1 Purpose and main conclusions

This document records the experimental directions explored in FRIS from June to October 2026, including work outside the scope of the main manuscript. It explains the questions behind the experiments, what was changed, what the results showed, and why the next direction was chosen. The evidence snapshot closes on 12 October 2026. It is a research record, not a revised paper or a claim that every explored method is ready for publication.

The work began by shaping illumination statistics and gradually separated three questions: what information reaches the detector, how accurately it is modelled, and how reconstruction uses it. The strongest extended results are learned 2D/3D PnP priors, joint changes to detection NA and sampling, and detector-aware training of a single phase plate. Their gains can combine, but not every combination or compression level improves every metric.

Five conclusions summarise the final state:

1. A phase encoder provides depth-distinguishing measurements in the evaluated geometry; a stronger learned prior does not remove the in-focus DMD's mirror ambiguity.
2. Learned 2D and 3D denoisers improve the PnP reconstruction. At the original detector, the learned solver at eight-fold compression approaches the generic-prior solver's full-measurement mean depth error.
3. Increasing detection NA together with detector sampling can replace many sequential illumination images. More pixels alone do not produce the same benefit.
4. Training a plate against detector-visible responses improves recovery at the original detector and adds to the learned-prior gain. At higher NA it improves mean depth placement but can reduce intensity fidelity.
5. Early illumination-statistics objectives, additional layers, alternative decoders and posterior sampling provided useful diagnostics, but were not consistently better reconstruction strategies.

The main manuscript remains unchanged. In particular, its training-agnostic reconstruction is a distinct operating point from the task-trained priors documented here.

The path through the experiments was not linear. Each stage changed the question asked next:

Table 1: Experimental progression and the questions that changed the next direction.

Period	Direction and turning point
June 2026	Test whether a cheap illumination-design objective could break mirror-depth symmetry.
September 2026	Correct the excitation physics, phase-feature scale and relay geometry before comparing encoders.
Late September	Find that improved field statistics and extra layers did not reliably improve recovery.
8 October	Restore fast voxel-integrated construction and settle on a single-plate reference.
9–10 October	Keep acquisition fixed and improve reconstruction with learned 2D/3D priors.
10–12 October	Study the detector, train against detector-visible signatures and consolidate the controls.

2 Experimental framework and reading conventions

2.1 Reference configuration

The later experiments use vectorial NA-0.8 excitation, linear fluorescence, a homogeneous medium and negligible specimen-induced scattering. Excitation and detection wavelengths are approximated as equal at 532 nm. A phase plate is propagated to an intermediate illumination plane and relayed into the specimen; the detection focus remains fixed while input angle changes.

Optical propagation uses a 384×384 grid at 0.08 μm. The analysis object is 32×96×96 at 0.32 μm, and reconstruction uses 16×48×48 voxels at 0.64 μm. The represented volume is 30.72×30.72×10.24 μm. The original detector has NA 0.1 and 12×12 samples per image. A single phase plate has 0.32 μm phase features and a 6.6 μm final gap. These are simulation and reconstruction sampling choices, not measured optical resolution.

The baseline reconstruction uses PnP-ADMM with the generic pretrained DnCNN-17M in three orthogonal slice orientations and a sparsity branch. The extended learned solver replaces the slice denoiser and adds a 3D denoising branch. Both use a 20-iteration reconstruction budget. Baseline bead tests in the original package used sparsity alone; the later learned-prior bead control instead compares PnP solvers using their neuronal-validation settings.

2.2 Measurements, noise and fair comparisons

Let P be detector samples per image, k illumination directions, $M = Pk$ scalar measurements and $N = 36{,}864$ reconstructed values. Measurement compression is $C = N/M$. At the original 12×12 detector, $k = 256, 64, 32, 16, 8$ corresponds to $C = 1, 4, 8, 16, 32$.

The final comparisons use the relative-Gaussian convention

$$q(M) = 15 + 5\log_{10}(N/M), \qquad \sigma(M) = \mathrm{mean}(\mathbf{y}_N^0)\, 10^{-q(M)/10}.$$

Here $\mathbf{y}_N^0$ is the relevant encoder/detector's noise-free reference stack at M=N. At a fixed detector this gives the square-root count scaling used in the main work. FRIS calls it the total-dose or equal-detected-photon convention. These names identify a normalisation assumption, not calibrated photon counts or a Poisson detector model. Earlier fixed-source-power and equal-illumination studies are labelled separately because their normalisation changes device rankings.

Image-count reduction and measurement compression must not be confused. A 48×48 detector with 16 images and a 12×12 detector with 256 images both supply 36,864 scalar measurements. The former uses sixteen times fewer images, but both have $C = 1$.

2.3 Test sets, tuning and evidence levels

The final quantitative test set contains eight prespecified mouse pyramidal neuron subvolumes with dendritic arbors: p64, p128, p192, p320, p448, p576, p704 and p832. Repeated noise realisations and illumination subsets are averaged within each subvolume before computing across-subvolume statistics. Subvolume 0 and the reference-selected qualitative examples p384/p973 are reported separately. PSNR, foreground SSIM and local depth error measure different aspects of recovery; depth error is a centroid-based metric, not axial resolution.

Extended solver and detector comparisons are tuned on held-out neuronal validation subvolumes p303, p10 and p693. This differs from the main manuscript's earlier vascular tuning. Each solver/encoder/detector/measurement setting is tuned independently, with boundary checks on the search grid. Consequently, extended baseline rows are not automatically interchangeable with the manuscript's baseline rows.

Evidence has three levels in this record. Final results are drawn from the consolidated 70-condition master table and paired comparisons. Earlier reconstruction results retain their original configuration and object labels. Historical surrogate scores describe what guided exploration; they are not final PnP results or measured resolution. The source index at the end identifies the relevant logs, scripts and exports.

3 Establishing a trustworthy simulation

3.1 From a proof of concept to shared excitation physics

The June implementation explored a decoupled illumination-design loss, avoiding differentiation through the full iterative reconstruction. It used a bimodal intensity target and axial/mirror decorrelation, initially with a scalar pupil-focusing model. Later objectives used exponential speckle statistics. Training reduced mirror correlation and showed that changing phase structure could break depth symmetry. This established a useful design direction, but not the final forward model.

The September physics audit found that the excitation paths did not share the same model: the imported DMD path used a vectorial PSF, whereas the phase-encoder path used scalar propagation, and the initial new coder used an objective-focusing geometry rather than the intended relay. FRIS replaced these with a common coherent vectorial relay. Intensity is formed after propagating the complex field and summing the squared magnitudes of its vector components; emission is modelled incoherently.

The audit also aligned the excitation focus with array index 15 on the 32-plane analysis grid. Mirror pairs must be defined about that focus rather than the midpoint of an even-length array. These corrections made encoder comparisons meaningful and superseded numerical claims from the earlier model.

3.2 Phase-feature scale and relay geometry

Early phase optimisation used 0.125 μm features. Re-evaluating the designed element on a 0.32 μm phase grid weakened its apparent mirror-separation advantage. The practical conclusion was to prevent training from relying on unresolved or non-propagating fine structure and to separate physical phase features from the finer numerical propagation grid. The 0.32 μm feature grid and 0.08 μm optical grid became the later reference.

The output-hop audit identified a 3.3 nm value in the inherited configuration, inconsistent with micrometre layer spacings. A 6.6 μm final gap was adopted after a gap study balanced angular-pattern changes against light leaving the finite window. The logged random two-layer study found near-angle correlation decreasing from 0.62 to 0.34 and 0.29 at gaps of 0, 6.6 and 10 μm, while relative retained energy fell from 1.00 to 0.73 and 0.63. These are historical two-layer diagnostics, not final one-plate recovery scores.

The implementation was later expressed as a phase element near an intermediate image plane. Fresnel scaling with relay demagnification 60 gives approximately 19 μm phase features, 24 mm defocus and 8–14 mrad element-side tilts. Numerical paraxial controls supported this interpretation; no hardware implementation was tested.

3.3 Fine measurements, integrated voxels and fast construction

Measurements are generated from a fine object; the reconstruction reference is its voxel average. They are not generated by simply replicating the coarse reference. The integrated operator is

$\mathsf{A} = \mathsf{BU}$, where B is the fine measurement process and U replicates each reconstruction value over its fine patch.

The FYP derivation was restored in the October implementation: a fine emitter's response is its illumination value multiplied by a reusable shifted emission footprint. Summing these products over a voxel patch constructs the integrated operator without repeatedly performing volumetric convolutions. The final one-plate build was logged at 109 s including preparation and validation. Live-forward checks were of order 8×10^{-7} relative error. A historical two-layer brute build took about 15 hours, but the configurations and timing boundaries were not matched, so their ratio is not a controlled speed-up measurement.

The retained patch-versus-coarse mismatch test showed 1.07% versus 25.7% forward discrepancy and 0.177 versus 1.859 μm depth error on subvolume 0. Direct optical simulation at 0.64 μm undersamples the high-NA field. This established that optical detail should be retained before integration onto the reconstruction basis. The auxiliary point-sampled arm is outside the accepted main comparison and is not needed for the conclusions here.

4 Illumination-design directions and what they taught us

4.1 Why illumination statistics were a tempting surrogate

The initial idea was to learn useful illumination without unrolling PnP. A surrogate is a cheaper objective used in place of the full reconstruction task. Here, the proposed losses included intensity-distribution matching, axial decorrelation, mirror separation, temporal decorrelation and phase smoothness. Different depths should produce distinguishable illumination signatures; the question was whether improving field statistics also improved the measurements retained after emission blur and detector averaging.

The important distinction emerged between raw illumination diversity and detector-visible information. A random complex field already achieved much of the desired fine-scale decorrelation. Improving a field statistic could therefore reduce the training loss without producing a useful advantage in the inverse problem.

4.2 Evolution of the objectives

Table 2: Historical illumination-design objectives and their practical outcomes.

Direction	Change tested	Outcome and next decision
Fine-plane decorrelation	Separate raw illumination planes; match a target intensity distribution	Random phase fields already performed well on these statistics; use reconstruction-scale and measurement-aware checks.
Voxel-scale depth decorrelation	Pool illumination and compare depth signatures	Mirror ambiguity was reduced, but sampled operator conditioning did not consistently beat random.
Single-plane distribution and temporal loss	Exponential PDF, temporal correlation, phase smoothness	Mean pattern correlation improved sharply, but near-angle redundancy and excessive contrast remained.
Near-angle and contrast guards	Gaussian angular weighting and per-pattern contrast penalty	Corrected those field statistics, but the weakest sampled operator direction still lagged random.
Loss-plane and smoothness changes	Remove phase smoothness; use pre-aperture or random axial planes; train longer	Small diagnostic changes, no consistent reconstruction advantage.
Additional layers and PDF removal	Five layers; two/five layers without the PDF term	More layers did not deliver the expected proxy-reconstruction gain; the PDF term had little effect on that score.
Output-characteristic loss	Retained light, voxel-scale contrast and axial decorrelation	Large changes in the intended characteristics, but little gain over efficient untrained diffusers.
Light-first loss	Increase the light-efficiency weight	More delivered light came with worse pattern quality and worse proxy reconstruction.

These variants were investigated mainly on physical two-layer configurations in September. Their scores should not be mixed with the October one-plate test-set results.

4.3 A diagnostic example: lower correlation was not enough

In the single-plane v2 experiment, mean pattern correlation fell from 0.204 for the random baseline to 0.024, while nearest-tilt correlation increased from 0.34 to 0.43 and contrast rose to 1.95. Contrast here is intensity standard deviation divided by mean intensity. The smallest singular value of a normalised 256-column sample fell from 0.0136 to 0.0034. The v3 near-angle/contrast correction improved that singular value to about 0.0100, still below random. Removing smoothness, changing the loss plane and increasing training duration did not close the gap reliably.

A centre-versus-edge diagnostic showed that removing peripheral columns did not remove the gap. The weak sampled direction involved separation of nearby depths around focus, rather than simply poor illumination at the field boundary. This shifted attention from one global conditioning number to noise-aware reconstruction diagnostics: a device should be judged by the recovery problem, not by whether one summary statistic improves.

Wiener-based surrogate reconstruction was then used to screen candidate encoders cheaply. Its rankings were checked against existing full reconstructions. It captured several encoder trends, but missed the designed-versus-random crossover at the lowest tested SNR and did not predict PnP's absolute PSNR. It became a screening tool, not a substitute for the final solver.

4.4 Normalisation, light efficiency and the limits of the early conclusion

Equal-mean illumination scoring removed differences in how much light passed through the aperture. A fixed-source-power proxy comparison instead preserved retained-light differences and initially favoured trained designs. Smoothed, untrained phase baselines then matched or exceeded those gains. In the historical 15-coder analysis, retained light strongly tracked the fixed-source-power proxy score, whereas pattern-to-pattern correlation tracked it weakly. These correlations motivated a light/contrast/axial loss, but optimising those characteristics did not reproduce a strong recovery gain.

An independent-speckle proxy produced nearly the same equal-illumination score as the random encoder, suggesting little room for improvement in that particular linear test. This did not prove that a random plate was globally optimal. The later detector-aware result changes that conclusion: field-statistics losses had failed, but training against the actual detector-visible voxel responses did improve recovery.

Conclusion: statistical field quality, average column correlation and one worst singular value are useful diagnostics, but none alone establishes recovery quality. Evaluate the proposed improvement through the measurement model and a held-out reconstruction test.

Evidence: decisions D8–D11o; `scratchpad/metric_new.py`, `validate_metric.py`, `metric_power.py`, `char_analysis.py` and `upper_bound.py`; corresponding `results/recon/` JSON records.

5 Architecture, scan allocation and noise studies

5.1 How many phase layers were needed?

A one-versus-two-layer experiment used 16 angularly spread directions, q=25, three random elements and three noise realisations. The neuronal example improved from 34.26 to 35.11 dB PSNR, with depth error decreasing from 0.74 to 0.50 reconstruction planes. Bead depth error and mirror leakage also decreased. One plate passed more light through the relay and was sufficient for the demonstrated depth encoding. This motivated the one-plate reference, rather than retaining extra layers merely because earlier training had used them.

The five-layer surrogate experiment was a different study: it accelerated some field-statistics improvements but did not improve the screened reconstruction score over matched random baselines. Together, these tests favoured a simpler acquisition architecture, not a claim that additional layers can never be useful.

5.2 Which angles, and how many spatial samples per image?

The FYP scan-allocation analysis favoured balanced azimuth/elevation sampling when judged by maximum pattern correlation. Direction-space geometry explains why equal angle steps are not equal phase-gradient steps. The later reference uses 16 azimuths and 16 signed elevations, covering an annulus of transverse directions. Short sequences taken from the beginning of scan order can concentrate along one azimuth, motivating random subsets across the full grid. First, grid, random, spread and Wiener-guided subset rules were implemented; the later decoder-screen summary reports less than 1 dB selection gains, not a robust universally best rule.

An earlier two-layer pixels-versus-images test held the scalar measurement count fixed. At M=2,304, 12×12 samples over 16 images gave 34.01 dB, while 6×6 over 64 gave 33.98 dB; coarser 4×4 and 3×3 sampling reduced PSNR to 32.78 and 32.47 dB. Modest binning could exchange spatial samples for more illumination patterns, but aggressive binning discarded useful lateral information. The later detection-NA study revisited this trade-off with changed optics and a stronger solver.

5.3 Noise sensitivity and validation boundaries

At the earlier q=15/25 conditions, neuronal recovery changed little with noise, so additional q=10/5 experiments tested a more noise-limited regime. With the one-plate baseline, full-measurement depth error on subvolume 0 was 0.177, 0.263 and 0.396 μm at q=15,10,5. Full PSNR was 37.22, 35.43 and 33.80 dB. At q=5, the 16-image result was 33.48 dB: acquisition compression added a relatively small PSNR cost once noise already limited recovery under the chosen count-dependent convention.

Several initial noise runs selected penalties at the edge of the tuning grid and produced misleading comparisons. Those rows were moved aside, the search ranges were expanded and the tests repeated. Bead validation also exposed how using the same bead positions for tuning and testing overstated recovery. A held-out phantom and stricter specimen-keyed tuning cache were adopted.

Conclusion: scan allocation and detector sampling interact with noise and regularisation. A tuning-boundary failure or reused validation object should not be interpreted as an optical result.

Evidence: 8 October history, D20–D21, `paper_results/package/layers/`, `package_1L/noise/one_plate/`, the original pixels-versus-images report and `compression_subsets.py`.

6 Learning a stronger reconstruction prior

6.1 The change: trained 2D slices plus a trained 3D branch

The extended solver track held the one-plate measurement model fixed and trained denoisers on reference volumes. The selected solver, t23n, uses a neuronal 2D DnCNN in all three slice orientations and adds a neuronal 3D DnCNN branch within PnP-ADMM. The 2D network has 17 layers and width 64; the 3D network has 12 layers and width 48. Both are trained with additive Gaussian corruption up to standard deviation 0.25 and learning rate 0.001. The saved runs used 30,000 iterations with batch 64 for the 2D model and 15,000 with batch 8 for the 3D model, taking about 31 minutes and 2.35 hours respectively. These are recorded run times, not a matched dimensionality benchmark. Weights are then frozen during reconstruction, and ADMM penalties are selected on held-out neuronal validation volumes.

This is not the training-agnostic operating point used in the main paper. It requires denoiser training and reference data, but does not train an end-to-end inverse network for each operator.

The result tests the future direction discussed in the manuscript rather than silently replacing its original method.

6.2 Recovery and compression range

The following table uses the final three-noise-realisation solver block, with repeats averaged within each of eight subvolumes. Lower depth error is better.

Table 3: Learned-prior comparison at the original detector; eight subvolumes, three noise realisations per subvolume.

Images / compression	Generic-prior depth (μm)	Learned-prior depth (μm)	Generic / learned PSNR (dB)
256 / C=1	0.497	0.346	31.62 / 33.20
64 / C=4	0.538	0.410	31.16 / 32.69
32 / C=8	0.647	0.503	30.61 / 32.11
16 / C=16	0.689	0.585	30.15 / 31.13
8 / C=32	0.811	0.820	29.71 / 30.00

The learned 32-image result has mean depth error close to the generic-prior 256-image result. The paired difference is $+0.006\pm0.091$ μm across subvolumes, with foreground-SSIM change -0.018 ± 0.056 and PSNR change $+0.49\pm2.00$ dB. This supports comparable mean depth placement at eight times fewer measurements in this test; it is not a formal equivalence test or a guarantee of identical feature recovery. The separate five-subset comparison gives $0.626\rightarrow0.476$ μm at 32 images. These different repeat sets are kept distinct.

Gains concentrate on sparse neurite-like volumes. Dense examples show little PSNR improvement, and one large crop-level gain raises the overall mean. The initial matched comparison's median PSNR gains were approximately 1.23, 1.00 and 0.86 dB at C=1,4,8. At C=32 the learned solver does not improve depth, despite a small mean PSNR increase. Training does not remove the information limit at very small measurement counts.

6.3 Leakage screen and controls

Because the stored references are crops of one parent stack without saved crop coordinates, training overlap was checked by an FFT shift/correlation screen. It excluded 141 of 1,013 candidate crops; 16 cleared crops were held out for validation, leaving 856 for training. A stricter re-screen found one genuine overlap just below the original volume threshold: training crop 1015 shares 1.99% with test crop p576. Excluding p576 left the learned-prior conclusion intact. This is a documented residual overlap, not a claim of a perfect leakage screen.

A vasculature-trained 2D/3D prior, tuned on the same neuronal validation volumes, gave full-measurement depth error 0.355 μm on the neuronal test set, close to the neuronal prior's 0.346 μm and below the generic prior's 0.497 μm. This supports a benefit beyond simply reproducing the neuronal training collection, although it is not an independent biological validation.

On synthetic beads, the neuronal prior produced zero false detections in the tested cases and Jaccard index 1.00. Depth error improved from 0.135 to 0.065 μm at full measurements and from 0.204 to 0.110 μm at 32 images. PSNR fell because the learned prior altered bead intensity profiles. Thus accurate localisation and intensity fidelity can move in opposite directions; these bead controls differ from the main paper's sparsity-prior bead results.

With an in-focus DMD, the learned solver still gave 1.226 μm depth error versus 0.346 μm with the phase plate at full measurements. The prior improved reconstruction but did not replace missing depth discrimination. A paraxial phase-plate control gave 0.354 μm with the learned solver, close to the reference angular-spectrum model.

Conclusion: learned volumetric priors are a useful compression lever in this collection, especially when combined with 2D slice priors. Their benefits depend on object structure and must be distinguished from the information supplied by the acquisition.

Evidence: D22; `solver_learned/README.md`, overlap-screen files and test rows; final master-table solver and encoder-control blocks.

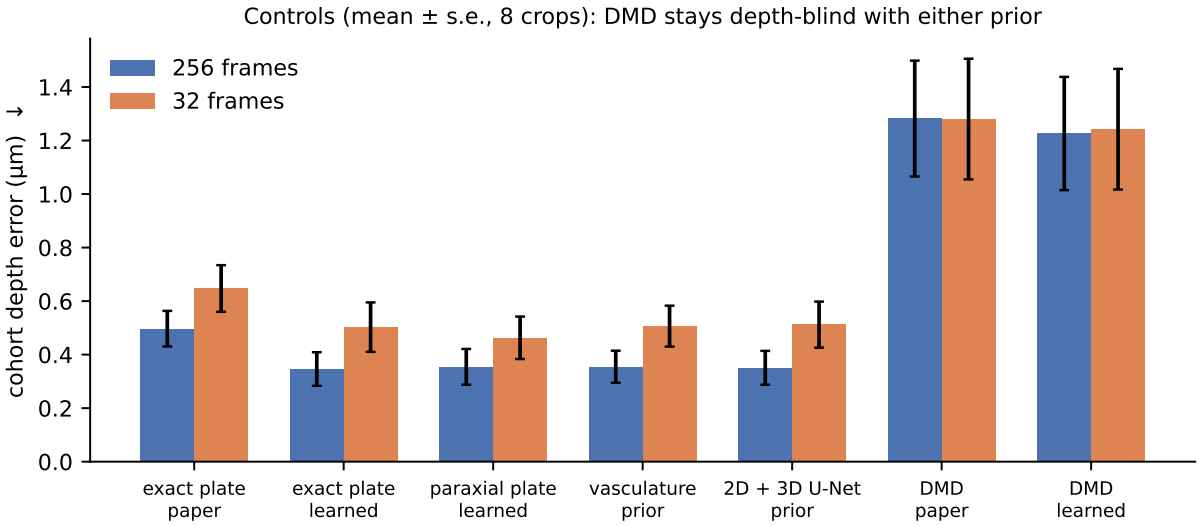


Figure 1: Encoder and prior controls. The same learned reconstruction remains depth-ambiguous with the in-focus DMD; the paraxial plate and vasculature-trained prior retain the phase-plate benefit. Bars compare 256 and 32 images; error bars are standard errors across eight test subvolumes. Original final-package plot.

7 Decoder alternatives and posterior sampling

7.1 Alternatives that did not replace the selected PnP solver

Table 4: Alternative reconstruction directions and why they were not adopted.

Direction	Observation	Conclusion for this study
Generic DnCNN variants	17L/17H/non-blind 17N did not improve the validation result	Changing generic noise variants was not sufficient.
Explicit non-negativity	Approximately baseline; iterates were already largely non-negative	No useful additional gain.
BM4D hybrid	Small validation gains, about 0.1–0.2 dB	A useful volumetric-prior clue, but not the best decoder.
Learned 2D or 3D alone	Smaller or inconsistent improvements	Combining learned slice and volume branches was stronger.
3D U-Net branch	Better standalone denoising, slightly worse PnP recovery	Denoiser quality alone did not rank inverse-problem performance.
Annealed non-blind HQS	No improvement over the selected solver	Not adopted.
MoDL-style unrolled 3D network	Two runs plateaued near 26 dB validation; output weakly depended on measurement count	Behaviour consistent with prior domination in this implementation, not a general failure of unrolling.
More ADMM iterations	100 iterations performed worse than 20 in the screen	Extra iteration count did not solve the compressed-recovery problem.
Independent random plate for every image	An acquisition oracle gained only about 0.15–0.45 dB in the partial validation screen	Removing the fixed-plate constraint offered little improvement in that test.

The MoDL experiment was checked with both fine-model and coarse-basis measurements. Similar performance in both cases argued against discretisation mismatch as the main cause of its failure. It therefore remained a decoder-training or measurement-use problem to investigate, rather than a reason to change the optical model. The tested alternatives were not adopted; their outcomes do not establish universal method rankings.

7.2 GPnP: estimation and uncertainty are different questions

Generative PnP was implemented as an annealed split-Gibbs sampler with non-blind 2D/3D denoisers and a perturb-and-optimise likelihood update using the stated noise scale. Sixteen chains were used for the final posterior-mean and uncertainty summaries. Its settings were calibrated on validation data for nominal 90% foreground coverage.

Table 5: GPnP estimation and foreground uncertainty; final eight-subvolume aggregation.

Images	GPnP mean PSNR (dB)	GPnP depth (μm)	Learned PnP PSNR (dB)	90% foreground coverage
256	33.11	0.363	33.20	0.606
64	26.93	0.785	32.69	0.807
32	26.19	0.903	32.11	0.782
16	23.88	1.066	31.13	0.766

At full measurements its mean estimate approached deterministic learned PnP, but compressed estimates were substantially worse. Predictive standard deviation correlated with absolute error at about 0.49–0.66, so the uncertainty maps contained useful information about where errors occurred. Their scale was less reliable: foreground intervals under-covered the reference, and the 16-image calibration selected the upper edge of its scale grid. High whole-volume coverage at some compressed settings did not imply good foreground calibration because most voxels were background.

Conclusion: GPnP did not improve compression in the tested implementation. It remains an uncertainty-analysis side result rather than the preferred estimator, and its intervals should not be treated as calibrated confidence statements.

Evidence: D23/D26; `solver_learned/` screening logs; final GPnP block and `tables/gpnp_uncertainty.csv`. The final eight-subvolume aggregation supersedes early ten-crop summaries that included showcase volumes.

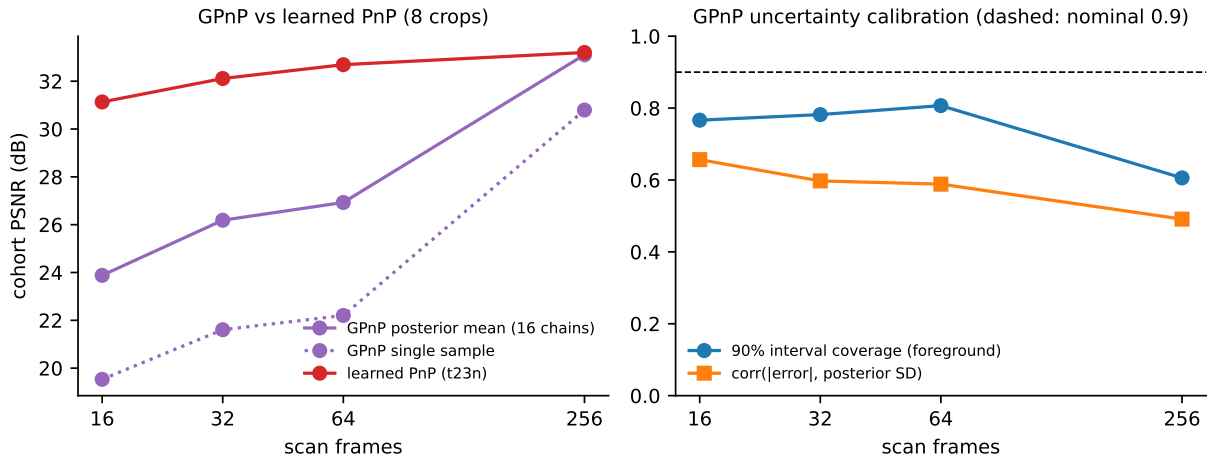


Figure 2: GPnP estimation and uncertainty. Posterior means deteriorate under compression, while nominal 90% foreground intervals under-cover the reference. Original final-package plot; sixteen chains and eight test subvolumes.

8 Detection NA and detector sampling

The next question was whether more information could be captured per illumination image. The detector study varied detection NA over 0.1, 0.2 and 0.3 and detector size over 12×12, 24×24 and 48×48. It kept the random phase plate and learned solver, tuned each configuration separately, and tested two fixed scalar-measurement counts: M=36,864 and M=4,608. Increased light collection from higher NA was not credited through a separate throughput advantage in the noise convention.

Table 6: Detection NA and spatial sampling at fixed scalar-measurement counts; random plate and learned prior.

Detection NA	Detector	Images at C=1 / C=8	Depth at C=1 / C=8 (μm)	PSNR at C=1 / C=8 (dB)
0.1	12×12	256 / 32	0.346 / 0.475	33.19 / 32.09
0.1	24×24	64 / 8	0.368 / 0.613	33.13 / 31.36
0.1	48×48	16 / 2	0.437 / 1.261	32.48 / 28.82
0.2	12×12	256 / 32	0.302 / 0.425	33.88 / 32.60
0.2	24×24	64 / 8	0.281 / 0.439	34.37 / 32.85
0.2	48×48	16 / 2	0.312 / 0.969	34.36 / 30.22
0.3	12×12	256 / 32	0.297 / 0.419	34.05 / 32.61
0.3	24×24	64 / 8	0.281 / 0.428	34.49 / 32.94
0.3	48×48	16 / 2	0.314 / 0.834	34.73 / 31.32

At C=1, NA 0.3 with a 48×48 detector used 16 images and improved both mean PSNR and depth error relative to the original detector's 256-image result. At C=8, the 24×24 high-NA configurations used eight images with mean depth error below the generic-prior baseline's full-measurement value. However, reducing to two images with a 48×48 detector degraded depth substantially.

The key is the combination of optics and sampling. At NA 0.1, substituting more detector pixels for illumination directions made recovery worse. Higher NA passes more lateral detail and changes defocus-dependent detector responses; a finer detector can retain this information. This explains the observed spatial–temporal trade-off without treating extra pixels as free measurement compression.

The first NA-0.3 full-measurement tuning ended on a penalty-grid boundary. A wider search replaced those rows; the final changes were at most about 0.1 dB. The consolidated table uses the corrected selections. A 96×96 option was not pursued because of footprint cost and limited expected benefit at the tested optical bandwidth.

Conclusion: more informative detection can replace many sequential images at fixed scalar-measurement count. The 16-image high-NA result is image-count reduction, not C=16 measurement compression.

Evidence: D24; final detector block, `detector_study/` rows, including the edge-corrected selections.

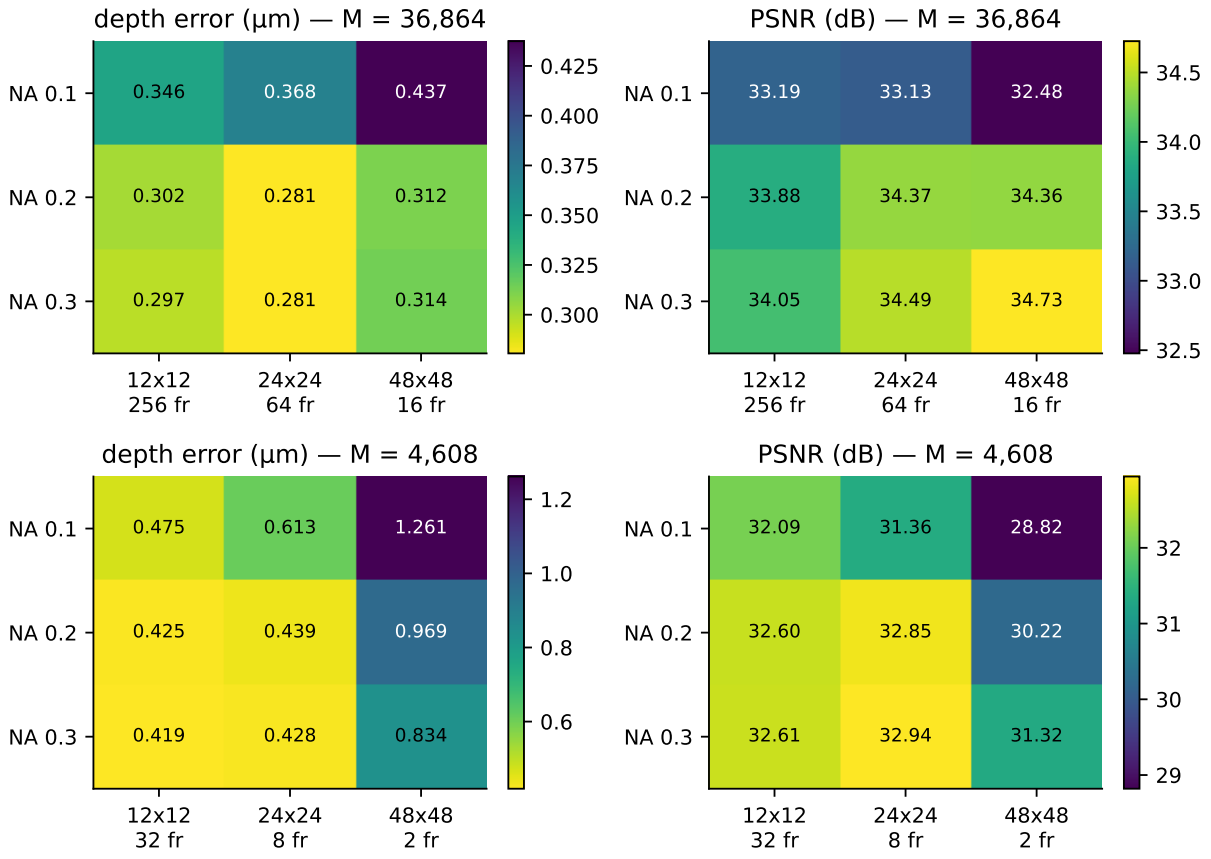


Figure 3: Joint detection-NA and sampling study with the random plate and learned prior. Within each row of panels the scalar measurement count is fixed, while camera-image count changes with detector size. Values are means over eight test subvolumes. Panels have separate colour scales; compare their printed values.

9 Detector-aware phase training and combined gains

9.1 Training the measurement signatures rather than raw speckle statistics

The final phase-training direction retained a single plate but changed the objective. Instead of optimising only illumination statistics, it formed voxel-integrated measurement vectors through the actual emission optics and detector. At sampled lateral positions, the 16 depth-voxel columns were centred and normalised, then their off-diagonal correlations were penalised:

$$\mathcal{L} = \mathbb{E}_{(x,y)} \left[\frac{\sum_{i \neq j} w_{ij} \left(\widetilde{\mathbf{a}}_i^{\top} \widetilde{\mathbf{a}}_j \right)^2}{\sum_{i \neq j} w_{ij}} \right].$$

Here $\widetilde{\mathbf{a}}_i$ is the centred, unit-norm measurement vector for depth voxel i at the sampled lateral position; w_{ij} doubles selected mirror pairs. Training used 600 Adam iterations, learning rate 0.05, 24 lateral positions and eight directions per step, starting from the reference random plate. The loss is scale-free and contains no specimen-specific reconstruction target. It is detector-aware training, not end-to-end training through the learned solver.

At NA 0.1/12×12, mean absolute depth correlation fell from about 0.855 to 0.543. At NA 0.3/48×48 it fell from about 0.83 to 0.50. The worst correlated pair remained near one, so training did not eliminate every ambiguous direction.

9.2 Original detector: a reproducible improvement

The table below uses the final detector/plate block, one noise realisation per subvolume and independently tuned solvers. It should not be substituted into the earlier three-realisation solver table.

Table 7: Random and trained plates at NA 0.1 with a 12×12 detector. Values are PSNR (dB) / depth error (micrometres).

Plate	Generic prior, C=1: PSNR / depth	Learned prior, C=1: PSNR / depth	Learned prior, C=8: PSNR / depth
Random 0	31.61 / 0.497	33.19 / 0.346	32.09 / 0.475
Random 1	31.58 / 0.495	33.17 / 0.354	32.10 / 0.506
Random 2	31.60 / 0.487	33.14 / 0.348	32.12 / 0.474
Trained	32.85 / 0.366	34.36 / 0.277	33.04 / 0.388
Trained repeat	—	34.44 / 0.279	33.05 / 0.383

PSNR is in dB and depth error in μm. Against each random plate, the trained plate improved PSNR in all eight subvolumes for both solvers at full and eight-fold-compressed measurements. Depth improved in all eight except two generic-prior compressed comparisons with seven improvements. A second training run reproduced the result. Random-plate variation was much smaller than the trained-versus-random mean gain.

Training also improved recovery with the generic prior, so the result was not dependent on the new learned decoder. Against random plate 0 at full measurements, the generic-prior paired depth change was -0.131 ± 0.068 μm; with the learned prior it was -0.069 ± 0.036 μm. This is the main successful validation of the detector-aware proxy against actual reconstruction.

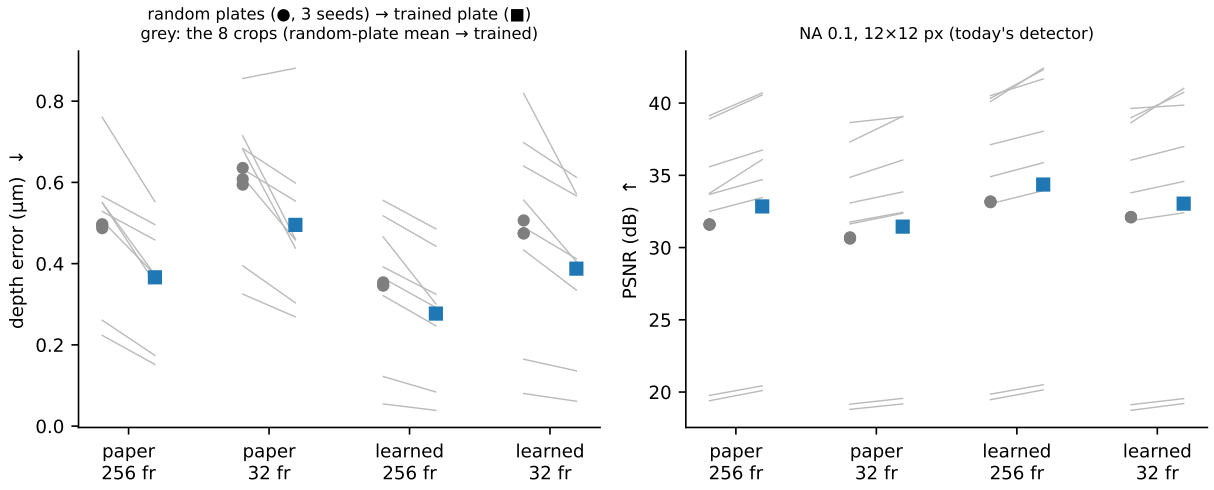


Figure 4: Detector-aware training versus three random plates at NA 0.1 and a 12×12 detector, with generic and learned priors at 256 and 32 images. Grey lines connect each subvolume's random-plate mean to its trained-plate result. Original final-package plot.

9.3 Higher NA: a mixed result and an untested direction

At NA 0.3/48×48, a plate trained for 16 images reduced mean depth error from 0.314 to 0.274 μm, improving six of eight subvolumes. Mean PSNR decreased by about 0.45 dB, largely because of one sparse crop. At two images, outside the training condition, depth error worsened from 0.834 to 1.176 μm. This is not a clean improvement at high NA, and it does not establish a successful two-image system.

9.4 Encoder and prior improvements can combine

At the original detector, trained plate plus learned prior with 32 images achieved 33.04 dB and 0.388 μm, compared with random plate plus generic prior with 256 images at 31.61 dB and 0.497 μm. The paired depth difference was -0.109 ± 0.100 μm, improving six of eight subvolumes; PSNR improved in five of eight. The mean advantage therefore stacks, but it is not a uniform per-subvolume win against that cross-compression reference.

Conclusion: detector-aware phase training succeeds where earlier field-statistics losses were inconclusive. The useful objective acts on the measurement signatures that reconstruction actually receives. Its success at the original detector does not automatically transfer to higher NA or fewer training directions.

Evidence: D25; `train_detector_aware.py`; final `designs/`, plate block and paired comparisons.

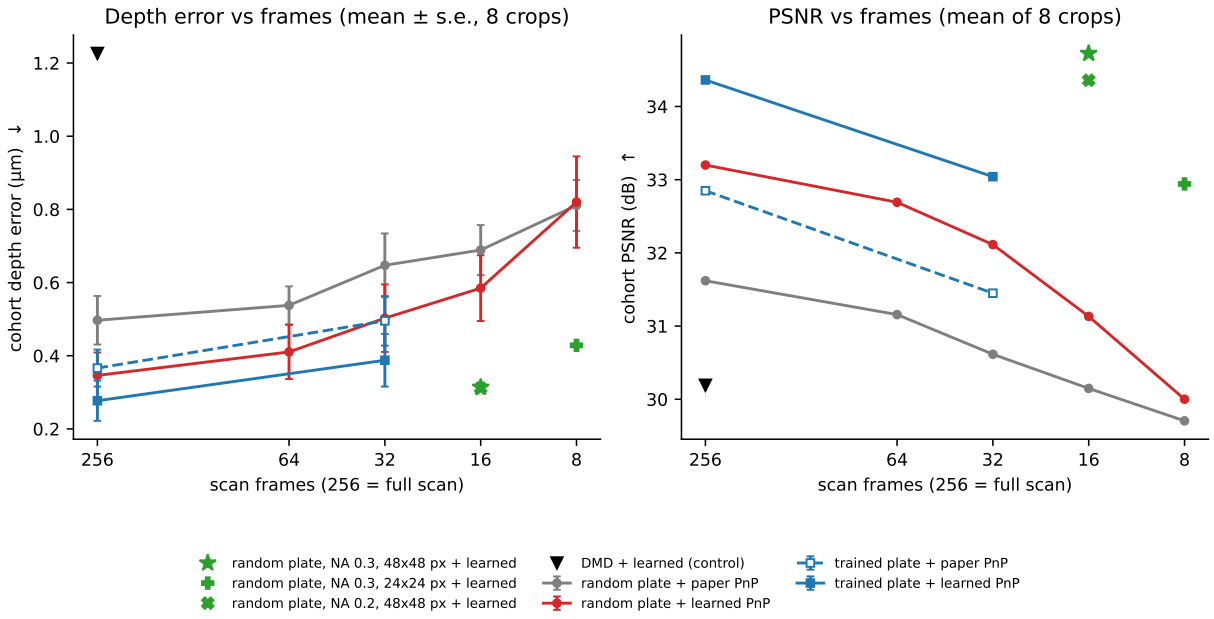


Figure 5: Observed recovery across encoder, prior and detector changes. The horizontal axis is camera-image count, not a common compression factor across detector sizes. Depth-error bars show standard errors across eight subvolumes, whereas tables in this record report means or sample standard deviations. This is an empirical comparison, not a proven optimal frontier.

10 Overall conclusions and directions left open

The experiments did not lead to a single universal optimisation rule. They showed where improvements could be made and why some promising directions did not carry through to recovery. Changes in configuration, normalisation, validation and metrics also explained several apparent contradictions.

The results identify three useful routes to improvement. Phase-structured illumination separates depth contributions, while fine-grid integration and the analytical construction make its measurement model practical. Reusing a 2D prior gives a simple training-agnostic reconstruction; learning 2D/3D priors improves the extended result. Finally, detection NA, spatial sampling and detector-aware phase training determine how much useful information each image contributes.

The most useful methodological lesson is to judge design changes at the detector and in held-out reconstruction, not solely by the appearance or correlation of illumination fields. A second lesson is to distinguish localisation, intensity fidelity and calibrated uncertainty: better depth error can coexist with worse PSNR, and uncertainty correlated with error can still under-cover the reference.

The following directions remained untested at the closing snapshot:

- A high-NA plate trained specifically for two to eight images, rather than reusing a plate trained for 16.
- Joint or end-to-end phase training through the selected learned PnP solver.
- Hardware calibration and experimental microscopy validation of the intermediate-image implementation.
- A calibrated photon-noise model, a direct optical-resolution study and uncertainty intervals that attain the intended foreground coverage.

These are recorded as open directions, not automatically requested experiments. The extended record can support a later manuscript decision without altering the current draft's scope or claims.

The completed directions should not be left labelled as future work in this record: task-trained 2D/3D PnP, higher-NA detector comparisons, detector-aware phase training and the paraxial recovery control all have results. What remains open is their further optimisation or experimental implementation.

11 Evidence index and reproducibility notes

All paths below are relative to `~/research/FRIS/`. Final numerical authority is `paper_results/final/`; historical conclusions are read in their dated configuration, with later corrections taking precedence.

Table 8: Source index for the extended experiments. Paths are relative to FRIS.

Evidence	Location
Chronology and decision rationale	`plan/history.md`; `plan/design_decisions.md`, D1–D27
Final aggregation and definitions	`paper_results/final/README.md`; `tables/master_table.csv`; `tables/paired_comparisons.csv`
Final uncertainty metrics	`paper_results/final/tables/gpnp_uncertainty.csv`
Generic-prior baseline, layers, noise, mismatch and paraxial bridge	`paper_results/package_1L/`
Learned priors, leakage screen, screening logs and qualitative controls	`paper_results/solver_learned/`
Encoder training and archived objectives	`coder/losses.py`; `scripts/coder/train_designed.py`; `scripts/coder/train_detector_aware.py`; `scratchpad/` diagnostics
Denoiser training and reconstruction	`scripts/denoise/train_denoiser.py`; `screen_overlap.py`; `pipeline/learned_denoiser.py`; `scripts/recon/solver_lab.py`
Detector and posterior experiments	`scripts/recon/detector_study.py`; `gpnp.py`; final raw-row folders
Integrated operator and numerical checks	`scripts/recon/build_fast.py`; `mismatch_ablation.py`; package operator metadata
Final result figures	`paper_results/final/figures/`; copied without scientific edits into this record

The final package is rebuilt by `scripts/recon/final_package.py` from saved CSV rows without running reconstruction. Its master table contains 70 conditions, each with eight test subvolumes. This document rounds selected means for readability; full precision, subvolume variation, repeat counts and per-condition source files remain in the CSVs. Original figure labels such as "paper PnP" identify the generic-prior solver, not a request to update the main manuscript; the extended controls use the validation protocol stated here.

Software recorded for the final server runs was SCICO 0.0.6, JAX 0.4.35, PyTorch 2.7.1+cu118 and Python 3.10.10. After a CUDA fault during concurrent work, the remaining runs used a serial GPU queue. No new experiment or remote job was launched to prepare this document.

11.1 Corrections that should not be lost

Earlier headings marked “active” can describe superseded architecture choices; the one-plate final state takes precedence over the initial two-layer default. Pre-vectorial and sub-feature-scale headline scores are historical and are not final evidence. Old field-statistics proxy claims of an optimal random plate are narrower than the later detector-aware results. The final GPnP summary excludes showcase crops and replaces older ten-crop averages. Different three-realisation, five-subset and detector-study tables should not be merged into one apparent condition. Crop p576’s residual training overlap remains documented, even though exclusion and cross-domain controls support the learned-prior conclusion.

Sections are in `contents/`. Compile `main.tex` to rebuild the PDF. The CSV tables and source notes are included in `data/`.